

Samantha Ajovalasit

Profile

Data Scientist with a PhD in Economics, Management and Statistics and a Master's in Big Data Analytics. Advanced in Python development for LLM deployment, model fine-tuning, and text processing pipelines using HuggingFace Transformers, spaCy, and scikit-learn on unstructured data at scale. Implementation and application of LLM models with agents (Qwen, Deepseek) via Ollama; full solution lifecycle from data ingestion through model training to deployment. Docker containerization, Git versioning, multidisciplinary team collaboration.

Experiences

2023 – 2026 **Researcher RTD (a)**, *University of Palermo*

Project *GRINS – Growing Resilient, Inclusive and Sustainable*. Development of pipelines in multiple environments for the analysis of complex socioeconomic data; statistical modelling and machine learning for decision support on public policies. Development of an interactive web-app (WhatD-SApp) integrated into the AMELIA cloud platform for public debt sustainability analysis.

2022 – 2023 **Research Fellow**, *University of Palermo*

Project *Just Smart – “Smart Justice: Tools and Models to Optimise Judges’ Work”*. Development of NLP algorithms for judicial text analysis. Data extraction and algorithmic evaluation to identify bottlenecks and propose data-driven solutions.

2021 – 2022 **Research Fellow**, *Ca’ Foscari University, Venice*

Project *H2020 Grant 101004488 European Media Platform – EUMEPLAT*. Development of NLP and machine learning analysis algorithms to explore sentiment patterns in European media. Contract Lecturer of Advanced Application of Computer Science for Cultural Heritage teaching basics of R and network analysis with interactive training and coding.

2017 **Internship**, *RJC Soft s.r.l., Pisa*

Project *Sprint*. Agent-based modelling and simulation of urban traffic patterns using Python and QGIS.

Education and Training

2018 – 2020 **PhD in Economics, Management and Statistics**, *University of Messina*, XXXIII Cycle, EQF 8

Thesis: “Disorientation towards routine immunization, COVID-19 anxiety, and propensity to vaccinate. Analysis based on Twitter data in Italy”. Defended: 21 December 2020.

Large-scale NLP analysis of Italian Twitter data: sentiment analysis, text classification and model fine-tuning for detecting vaccination hesitancy and measuring the impact of political communication on vaccination propensity. Use of spaCy, scikit-learn and supervised machine learning on Italian text corpora.

2017 – 2018 **II Level Master's in Big Data Analytics & Social Mining**, *University of Pisa*, EQF 7

Thesis: *Traffic Analysis in Urban Environments via GPS Trajectories*. Defended: 4 May 2018. Covers data mining, machine learning, data analysis & visualization, network science, computational social science, ethics, data journalism. Focus on big data management, knowledge extraction, and decision support - (Python, R, SQL, Knime).

2013 – 2016 **Master's Degree in Economics**, *University of Pisa, Scuola Superiore Sant'Anna*, EQF 7
Advanced economics training with quantitative and statistical tools. Analyzes economic phenomena at firm, industry, national, and international level. Covers econometrics theory and software: Gretl, R, Matlab. Thesis: *Option Pricing with Monte Carlo Simulation and Variance Reduction Technique*. Simulation of Option Pricing with Monte Carlo and Pseudo Monte Carlo methods to reduce the variance of the simulations. Defended: 5 July 2016.

2007 – 2012 **Bachelor's Degree in Economics and Finance**, *University of Palermo*, EQF 6
Thesis: *"Loan Transfer and Securitisation"*, Defended: 5 November 2012.

Technical Skills

LLM & Generative AI	Deployment via Ollama (LLaMA, Mistral, Gemma, Deepseek). Fine-tuning on text corpora. Prompt engineering, production pipeline integration, full model lifecycle management.
NLP & ML	spaCy, HuggingFace Transformers, scikit-learn, NLTK. Text classification, sentiment analysis, Italian judicial NER. Data extraction, annotation (Doccano), web scraping (Selenium).
Big Data & Data Engineering	Spark, Hadoop, Hive. SQL/MySQL, Postgres + PostGIS. ETL pipelines, end-to-end architecture. NoSQL (MongoDB).
Web App & Visualization	Full-stack: Bootstrap 4, HTML/CSS, Plotly/Dash, Shiny (R). Drag-and-drop upload, real-time display. WhatDSApp, EUMEPLAT Dashboard.
Quantitative Finance	Monte Carlo with variance reduction. Stochastic Debt Sustainability Analysis (SDSA). ESG modeling, portfolio optimization, Option Pricing. Financial time series analysis.
Programming Languages	Python (advanced), R, SQL, MATLAB. REST APIs, JSON. Git/GitHub, Docker containerization.
Systems & Infrastructure	Linux (Arch, Ubuntu), cloud platforms (AMELIA). VS Code Remote SSH, shell scripting. Sysadmin in computational research.
Languages	Italian (native), English (B2), French (B1).

Software and Tools Released

WhatDSApp	Interactive web-app for public debt assessment. Integrates sustainability analysis to create optimal financing scenarios and strategies. AMELIA platform, GRINS project: https://ameliadp.grins.it/
EUMEPLAT	Interactive dashboard of the data clustering reports. EUMEPLAT Dashboard

Additional Training

- 2019 **Summer School in Data Science and Epidemic Models**, *University of Trento*, Complex data analysis, digital tracking, human mobility, computational methods and social media analytics.
- 2019 **Fundamental of Digital and Computational Demography**, *Max Planck Institute for Demographic Research*, Rostock, Big data analytics for demographic research and social science.
- 2018 **Summer School on Clustering, Data Analysis and Visualization of Complex Data**, *University of Catania*, Clustering, classification, visualisation of large datasets; numerical analysis and cloud-computing platforms.

Teaching Activities

- 2023 – 2026 **Lecturer for the course *Mathematics for Business***, Bachelor's Degree in Economics and Business Administration – University of Palermo., Calculus and Financial Mathematics
- 2018 – 2020 **Senior Tutor**, University of Catania, Course: *Data Science and Demographic Processes and Economic Statistics*. NLP, sentiment analysis and REST API integration for Twitter data analysis

Proceedings, Papers and Publications

- 2025 **Are bad governments threat to sovereign defaults? Effects of political risk on debt sustainability.** Working Paper 01/2025, Bruegel. Bruegel.org
- 2025 **How much of threat to US debt sustainability is Trump's One Big Beautiful Bill Act?** Analysis Report. Bruegel. Bruegel.org
- 2024 **Debt Sustainability in Context of Population Ageing: Risk Management Approach.** Risks 12(12). <https://www.mdpi.com/2227-9091/12/12/188>
- 2021 **Evidence of disorientation towards immunization on online social media.** PlosOne. <https://doi.org/10.1371/journal.pone.0253569>

Scientific Society Memberships

AMASES, Italian Association for Mathematics Applied to Social and Economic Sciences. Member since 2023.